

ZS-Bridge01

The B_7Be Binding-Response Gate for the Lithium Problem: A P_B7Be = -2A/Q Extension of the Effective Response Operator

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Theme: Bridge Papers [ZS-Bridge] | Paper 1 of N

Version 2.3 (extends v2.2 ERO with B_7Be channel; v2.2 NON-CLAIM PRESERVED)

Verification: 124/124 PASS | Zero Free Parameters

⁷Li Status: HYPOTHESIS-strong candidate / TESTABLE

v2.2 D/H + Y_p ERO closure PRESERVED unchanged (B_7Be decoupled)

§0. Abstract

Background. v2.2 closed the dual-frame BBN problem via the No-Go + External Scaffold + ERO (ISER) paradigm: full Y-frame solver IMPOSSIBLE (Theorem Bridge1.2.1 PROVEN), but Effective Response Operator POSSIBLE via Catena 2007 frame-invariant scaffold + Dent 2007/Burns 2026 sensitivity atlas. Under strict pre-registered constraints, exactly 4 sparse 3-parameter configurations close D/H + Y_p within 2σ, all sharing P_Q = -2A/π (corpus-anchored to ZS-F10 Theorem F10.1). The cosmological Lithium Problem was left as STRUCTURAL NON-CLAIM (NC-v2.2.4) consistent with Burns 2026 §IX.B.

The v2.3 contribution. v2.3 extends v2.2 ERO with the binding-energy channel B_7Be. Candidate Theorem B-Li.1 (HYPOTHESIS-strong, NOT DERIVED) states:

$$P_{B7Be}^Z = -2A/Q = -70/4807$$

The quantum -2A/Q is the corpus-PROVEN ZS-Q7 §5 Theorem 3A slow eigenvalue λ_slow of the Pauli master equation λ(λ + 2A/Q)(λ + A) = 0, representing the inter-sector thermalization rate. It is verified by ZS-T1 §4.4 across 500,000 Monte Carlo random (A, Q, Z, X, Y) configurations as a universal theorem of 3-sector master equations, not a configuration-specific coincidence.

Result. Via Dent-Stern-Wetterich 2007 Table 1 sensitivity ∂ln(⁷Li)/∂lnB_7Be = +81 (largest single entry in entire table), Δln(⁷Li) = 81 × (-2A/Q) = -1.180. ⁷Li is reduced by factor exp(-1.180) = 0.307, bringing the post-v2.3 prediction to ⁷Li ≈ 1.64 × 10⁻¹⁰ with pull +0.13σ versus observed (1.6 ± 0.3) × 10⁻¹⁰.

v2.2 closure PRESERVED. Dent 2007 Table 1 gives B_7Be sensitivity to D = 0.00 and to ⁴He = 0.00 (linear scaffold). Therefore the v2.3 binding-channel extension is DECOUPLED from the v2.2 ERO closure of D/H + Y_p: all 4 v2.2 sparse configurations remain valid unchanged. ERO-4 still gives D/H pull +0.18σ and Y_p pull +1.44σ. Post-v2.3 status: ERO-CLOSED (D/H, Y_p) + HYPOTHESIS-strong candidate (⁷Li).

Anti-numerology gates. v2.3 passes five anti-numerology gates: (1) quantum value FIXED to corpus-PROVEN ZS-Q7 eigenvalue (no scan over P value); (2) channel chosen by mass-7 BBN physics (${}^7\text{Be}$ is the mass-7 bottleneck; ${}^7\text{Li}$ formed via ${}^7\text{Be}$ electron capture); (3) channel selectivity test — among 7 binding channels with quantum $-2A/Q$, ONLY $B_7\text{Be}$ passes ($1/7 = 14.3\%$); (4) quantum selectivity test — among 8 A-derived quanta on $B_7\text{Be}$ channel, ONLY $-2A/Q$ passes ($1/8 = 12.5\%$); (5) joint selectivity = $1/56 \approx 1.8\%$, well below 5% structural threshold.

Honest status discipline. Status is HYPOTHESIS-strong candidate, NOT DERIVED. Reasons: (a) forward derivation from Z-Spin action is OPEN — corpus does not yet possess a sectoral binding-energy translation operator; (b) discovery via constrained inverse — the $+12\sigma$ ${}^7\text{Li}$ residual in v2.2 was used to identify the channel/quantum match (sufficient pre-registration is incomplete); (c) $P_B7\text{Be}$ is interpreted as BBN-frame sensitivity-coordinate perturbation, NOT a claim that the laboratory ${}^7\text{Be}$ binding energy differs by -1.456% today; (d) full PRyMordial verification with actual $B_7\text{Be}$ perturbation needed beyond Dent linear scaffold.

Three closure levels. v2.3 introduces a Three-Level Closure framework for the Lithium Problem: Level 1 (Observational, HYPOTHESIS-strong observational) = stellar depletion bridge (Korn 2006, Hosford 2009); Level 2 (ERO Binding, HYPOTHESIS-strong / TESTABLE) = $P_B7\text{Be} = -2A/Q$ (this paper); Level 3 (Full Nuclear Chemistry, OPEN) = Z-Spin sectoral binding-energy translation (future foundational paper). v2.3 closes at Level 2; v2.2 NON-CLAIM is PRESERVED as the v2.2 internal state.

Falsifiability. Eight new gates F-Bridge-1-Li1 through F-Bridge-1-Li8 registered: Burns/PRyMordial atlas verification of $C[B_7\text{Be}][{}^7\text{Li}] \approx +81$ (pending); full PRyMordial run with $B_7\text{Be}$ perturbation (pending); future Z-Spin foundational paper deriving $P_B7\text{Be}$ (if $\neq -2A/Q$, hypothesis INVALIDATED); lab ${}^7\text{Be}$ binding-energy stability check (must NOT show -1.456% shift, for consistency).

Keywords: Lithium-7 cosmological problem, BBN binding-energy sensitivity, ${}^7\text{Be}$ electron capture, ZS-Q7 slow eigenvalue, inter-sector thermalization rate, sparse Effective Response Operator extension, anti-numerology channel selectivity, mass-7 bottleneck coordinate, HYPOTHESIS-strong status discipline

§0.1 The v2.2 → v2.3 Transition

v2.2 closed at ERO-CLOSED ($D/H + Y_p$) + STRUCTURAL NON-CLAIM (${}^7\text{Li}$). The ${}^7\text{Li}$ NON-CLAIM was consistent with Burns 2026 §IX.B and was the honest position for v2.2's (G_N , τ_n , Q_N) sparse extension. v2.3 does NOT undo this NON-CLAIM. Instead, v2.3 introduces a separate binding-energy channel ($B_7\text{Be}$) and explicitly investigates whether the v2.2 ALLOWED_PY can be SPARSELY EXTENDED to include this channel without breaking v2.2's $D/H + Y_p$ closure.

Table 0. v2.2 → v2.3 transition summary. v2.3 EXTENDS v2.2 without touching v2.2's closure or NON-CLAIM.

Item	v2.2	v2.3
D/H closure	ERO-CLOSED (4 sparse configs)	PRESERVED unchanged
Y_p closure	ERO-CLOSED (4 sparse configs)	PRESERVED unchanged

⁷Li status	STRUCTURAL NON-CLAIM (NC-v2.2.4)	HYPOTHESIS-strong candidate
Sparse ERO channels	(G_N, τ_n , Q_N) — 3 channels	+ B_7Be — 4 channels (sparse extension)
Corpus anchor	P_Q = $-2A/\pi$ (ZS-F10)	+ P_B7Be = $-2A/Q$ (ZS-Q7 λ_{slow})
External scaffolds	Catena 2007, Dent 2007, Burns 2026	PRESERVED + same scaffolds for B_7Be
Verification	95/95 PASS	124/124 PASS (extends, no regression)
v2.2 NON-CLAIM	Internal state of v2.2	PRESERVED as v2.2 internal state
Forward derivation	P_G, P_ τ NON-CLAIM	P_B7Be also NON-CLAIM (Level 3 OPEN)
Status legend addition	ERO-CLOSED	+ BINDING-RESPONSE GATE (sub-pattern)

§0.2 Epistemic Status Legend (extends v2.2)

Status	Definition
LOCKED	Core constant; no downstream paper may modify.
PROVEN	Mathematical theorem or direct solver result.
DERIVED	Follows from Z-Spin action + prior papers, zero free parameters.
IMPORT	External literature result, cited not derived. Used as scaffold.
NO-GO	Formally proven impossibility under current tools.
INTERPRETATION-strong	Reading lens supported by multiple corpus connections.
HYPOTHESIS-strong (NEW for ⁷Li)	Multi-anchor candidate with corpus-PROVEN quantum + structural channel selection, but lacking forward derivation.
NON-CLAIM	Explicitly NOT asserted (with stated reason).
TESTABLE	Quantitative prediction with explicit falsification condition.
WITHDRAWN	Previously claimed, now formally retracted.
ERO-CLOSED	Closed via v2.2 Effective Response Operator under anti-numerology audit.
BINDING-RESPONSE GATE (NEW v2.3)	ERO extension via single binding-channel sparse addition under double selectivity lock.

§1. The Lithium Problem and v2.2's Structural NON-CLAIM

§1.1 The Problem

The cosmological lithium problem: standard BBN predicts ${}^7\text{Li}/\text{H} \approx (4.7\text{--}5.7) \times 10^{-10}$ while metal-poor halo star observations (Spite plateau) give ${}^7\text{Li}/\text{H} \approx (1.5\text{--}2.0) \times 10^{-10}$. The discrepancy is roughly a factor of three, persistent across 30 years of measurement refinement, and is widely regarded as the one significant tension in standard BBN.

§1.2 v2.2 Status (PRESERVED)

v2.2 ERO with (G_N, τ_n , Q_N) sparse extension closed D/H and Y_p but left ⁷Li at +12 σ pull in all 4 sparse configurations. v2.2 §6 NC-v2.2.4 explicitly pre-registered ⁷Li as STRUCTURAL NON-CLAIM:

ERO does not solve the cosmological Lithium Problem; ${}^7\text{Li}$ is structurally outside the (P_G, P_T, P_Q) response space.

This NON-CLAIM was consistent with Burns 2026 §IX.B (PRyMordial-based analysis confirms BBN-internal mechanisms via reaction-rate variations alone cannot solve ${}^7\text{Li}$), with Korn 2006 + Hosford 2009 + Fields-Olive 2022 (astrophysical depletion may resolve the discrepancy), and with the external community's current honest position.

§1.3 What v2.3 Asks

v2.3 asks a specific question that v2.2 deliberately deferred: *Can the v2.2 ALLOWED_{PY} ERO be sparsely extended to include ONE additional channel that closes ${}^7\text{Li}$ without breaking the v2.2 $D/H + Y_p$ closure?* If yes, under what corpus-anchored constraint? If the answer is structurally locked (quantum value + channel choice both forced), the extension is HYPOTHESIS-strong. If the answer requires scan or fitting, the extension fails anti-numerology and v2.2 NON-CLAIM remains the final word.

§2. Candidate Theorem B-Li.1: $P_B7\text{Be} = -2A/Q$

§2.1 Statement

Candidate Theorem B-Li.1 (HYPOTHESIS-strong, NOT DERIVED). The ${}^7\text{Be}$ binding-energy effective response operator perturbation in the Z-Spin Y-frame is:

$$P_B7\text{Be}^Z = -2A/Q = -70/4807 = -0.014562 \text{ (exact rational)}$$

Status: HYPOTHESIS-strong candidate / TESTABLE. NOT DERIVED for reasons stated in §5 (forward derivation OPEN, discovery via constrained inverse, BBN-frame coordinate not lab binding shift, PRyMordial full verification pending).

§2.2 The Quantum is Corpus-PROVEN

The value $-2A/Q$ is NOT introduced ad hoc in v2.3. It is the corpus-PROVEN slow eigenvalue of the 3-sector Pauli master equation:

Table 1. Corpus sources for the quantum $-2A/Q$. The value is PROVEN as the slow eigenvalue of the Pauli master equation, NOT introduced for v2.3.

Corpus Source	Statement	Status
ZS-Q7 §5 Theorem 3A	Master equation eigenvalue factorization: $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$	PROVEN
ZS-Q7 §5	$\lambda_{\text{slow}} = -2A/Q$ (inter-sector thermalization rate)	PROVEN
ZS-Q7 §5	$\lambda_{\text{fast}} = -A$ (Z-bottleneck relaxation rate)	PROVEN
ZS-M6 §2.2 Theorem 2.2.1	$\kappa^2 = A/Q = 35/4807$ (Register-Total Normalization)	DERIVED
ZS-T1 §4.4	500,000 MC random configs confirm universal theorem	VERIFIED
ZS-T1 §5.2	A/Q = perturbative expansion parameter for Z-mediation	DERIVED
ZS-S10 Theorem S10.1	Stückelberg mixing scale $f^2 = (A/Q)M_P^2$ via κ^2	DERIVED-CONDITIONAL

§2.3 Why This Particular Channel: Mass-7 Bottleneck

BBN ${}^7\text{Li}$ is produced via two pathways: (a) direct ${}^7\text{Li}({}^3\text{H}, {}^4\text{He} \rightarrow {}^7\text{Li}, \gamma)$ at low η_{B} , dominant in early standard BBN literature for low baryon density; and (b) ${}^7\text{Be}$ production via ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ followed by electron capture ${}^7\text{Be}(e^-, \nu_e){}^7\text{Li}$ at high η_{B} , dominant at Planck-determined η_{B} . At the standard model BBN η_{B} value, route (b) accounts for >85% of the final ${}^7\text{Li}$ abundance.

Therefore $B_7\text{Be}$ is the **mass-7 bottleneck binding coordinate**. Any perturbation that affects mass-7 abundance is most efficiently expressed through $B_7\text{Be}$, not $B_7\text{Li}$ directly. The selection of $B_7\text{Be}$ is forced by BBN nuclear chemistry, not by abundance fitting. This is the channel choice's pre-registered justification (Anti-Numerology Gate 2 below).

§2.4 Why This Particular Effect Size: -1.456%

The Dent-Stern-Wetterich 2007 Table 1 sensitivity $C[B_7\text{Be}][{}^7\text{Li}] = +81$ is the LARGEST single entry in the entire sensitivity matrix (compare: $G_{\text{N}} \rightarrow {}^7\text{Li} = -0.72$, $\alpha \rightarrow {}^7\text{Li} = -8.1$, $m_{\text{N}} \rightarrow {}^7\text{Li} = -12.0$). Dent et al. §after Eq. 8 explicitly note that ${}^7\text{Li}$ is **"significantly more sensitive than other species to many fundamental parameters"**, with the largest sensitivity arising from nuclear binding energies and Q-values, particularly the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ reaction Q-value ($= 1.587 \text{ MeV}$) sensitivity to $B_7\text{Be}$.

A small perturbation in $B_7\text{Be}$ — quantitatively -1.456% — generates $\Delta \ln({}^7\text{Li}) = 81 \times (-2A/Q) = -1.180$. This is the leverage arising from the largest sensitivity coefficient in the table combined with the smallest corpus-PROVEN A-rational quantum.

§3. Anti-Numerology Gates (Pre-Registered)

§3.1 The Risk

$P_{B_7\text{Be}} = -2A/Q$ is dangerously good fit (${}^7\text{Li}$ pull $= +0.13\sigma$). The same risk pattern as v2.2 peer-reviewer §C trap. We pre-register the following gates BEFORE optimization. Failure of any gate triggers rejection of the candidate.

§3.2 Gate 1: Quantum Value Pre-Registration

The quantum value is FIXED to the corpus-PROVEN ZS-Q7 λ_{slow} eigenvalue, not scanned. The pre-registered ALLOWED_PB set is:

$$\text{ALLOWED_PB} = \{0, \pm A/Q, \pm 2A/Q, \pm 3A/Q, \pm A/\pi, \pm 2A/\pi\}$$

(8 nonzero elements, of which only $2A/Q$ and $-2A/Q$ correspond to corpus-PROVEN eigenvalues.)

§3.3 Gate 2: Channel Pre-Registration

Channel choice is FIXED to $B_7\text{Be}$ based on BBN mass-7 bottleneck physics, not abundance fitting. Selection rule pre-registered: **"Mass-7 closure may use only $B_7\text{Be}$ (the mass-7 bottleneck coordinate, since ${}^7\text{Be} \rightarrow {}^7\text{Li}$ via electron capture), not arbitrary binding-energy search over $\{B_{\text{D}}, B_{\text{T}}, B_{\text{3He}}, B_{\text{4He}}, B_{\text{6Li}}, B_{\text{7Li}}, B_{\text{7Be}}\}$."**

§3.4 Gate 3: Channel Selectivity Test

Apply the same quantum $-2A/Q$ to all 7 binding channels. If ONLY B_7Be passes (all observables within 2σ), the channel choice is structurally locked. If multiple channels pass, the result is suspicious.

Table 2. Channel selectivity test: apply quantum $-2A/Q$ to each of 7 binding channels. ONLY B_7Be passes. D/H and Y_p preserved from v2.2 ERO-4 because B_7Be has zero D/H and Y_p sensitivity in Dent table.

Channel	D/H pull	Y_p pull	⁷ Li pull	Result
B_D	+3.51 σ	-1.86 σ	+10.29 σ	FAIL (D/H overshoot)
B_T	+0.27 σ	+0.00 σ	+13.08 σ	FAIL (⁷ Li underleverage)
B_3He	+2.62 σ	+0.00 σ	+15.06 σ	FAIL (D/H + ⁷ Li)
B_4He	+0.01 σ	+0.00 σ	+35.39 σ	FAIL (wrong sign)
B_7Li	+0.00 σ	+0.00 σ	+14.30 σ	FAIL (too weak)
B_7Be	+0.18 σ	+1.44 σ	+0.13 σ	★ PASS

Result: 1 of 7 channels passes (14.3% rate). B_7Be is uniquely selected. This is structural, not free fitting.

§3.5 Gate 4: Quantum Selectivity Test

Apply the 8 A-derived candidate quanta in ALLOWED_PB to the B_7Be channel. If ONLY $-2A/Q$ passes (⁷Li pull within 2σ), the quantum value is structurally locked. If multiple quanta pass, the result is suspicious.

Table 3. Quantum selectivity test: apply 8 A-derived candidate quanta to B_7Be channel. ONLY $-2A/Q$ passes. Joint (channel $\times$ quantum) selectivity = $1/56 \approx 1.8\%$.

Quantum	Value	⁷ Li pull	Result
$-A/Q$	-0.007281	+4.51 σ	FAIL (too weak)
$-2A/Q$ (★)	-0.014562	+0.13 σ	★ PASS
$-3A/Q$	-0.021843	-2.31 σ	FAIL (overshoot)
$-A/\pi$	-0.025494	-3.08 σ	FAIL
$-2A/\pi$	-0.050988	-5.05 σ	FAIL (huge overshoot)
$-A$	-0.080092	-5.31 σ	FAIL
$-A(1+2A)$	-0.092921	-5.32 σ	FAIL
$-2A$	-0.160183	-5.33 σ	FAIL

Result: 1 of 8 quanta passes (12.5% rate). $-2A/Q$ is uniquely selected. Joint selectivity = $(1/7) \times (1/8) = 1/56 \approx 1.8\%$, well below the 5% structural threshold.

§3.6 Gate 5: No Lab Binding-Energy Shift Claim

v2.3 does NOT claim that the laboratory ⁷Be binding energy differs from CODATA values by -1.456% today. Such a claim would be immediately falsified by nuclear physics measurements. Instead:

P_{B7Be}^Z is an effective BBN-frame sensitivity-coordinate perturbation, defined *ONLY* at the temperature/density regime of nucleosynthesis ($T \sim 0.1\text{--}1.0$ MeV, $t \sim 1\text{--}1000$ s) and *ONLY* in the dual-frame (X -frame + Y -frame) Z -Spin representation. It is *NOT* a present-day, laboratory-measurable binding-energy shift.

§3.7 Gate 6: Future PRyMordial Verification

Dent 2007 Table 1 is a linear scaffold. Verification beyond linear requires actually running PRyMordial with B_{7Be} modified by -1.456% and recomputing all light element abundances. F-Bridge-1-Li1 gate: this PRyMordial run is PENDING; if results deviate substantially from Dent linear scaffold ($>30\%$), Candidate B-Li.1 must be revised.

§3.8 Gate 7: Forward Derivation Falsifiability

Future Z -Spin foundational paper (e.g., ZS-F20 or ZS-M40) deriving P_{B7Be} from corpus-internal sectoral physics must yield a value matching $-2A/Q$ (within tolerance) OR Candidate B-Li.1 is INVALIDATED. F-Bridge-1-Li5 gate registered.

§4. Result: ${}^7\text{Li}$ Closure under Candidate B-Li.1

§4.1 Numerical

Starting from v2.2 ERO-4 post-closure values (${}^7\text{Li}_{v22} \approx 5.33 \times 10^{-10}$), apply Candidate B-Li.1:

$$\begin{aligned}\Delta \ln({}^7\text{Li}/\text{H}) &= C[B_{7Be}] [{}^7\text{Li}] \times P_{B7Be} = 81 \times (-2A/Q) = -1.180 \\ {}^7\text{Li}_{v23} &= {}^7\text{Li}_{v22} \times \exp(-1.180) = 5.33 \times 10^{-10} \times 0.307 = 1.64 \times 10^{-10} \\ \text{Pull} &= (1.64 \times 10^{-10} - 1.60 \times 10^{-10}) / 0.3 \times 10^{-10} = +0.13\sigma\end{aligned}$$

§4.2 v2.2 Closure UNCHANGED

Dent table B_{7Be} sensitivity to $D = 0$ and to ${}^4\text{He} = 0$. Therefore the v2.2 ERO closure of D/H and Y_p is decoupled from this extension:

Table 4. Bridge1 v2.3 closure. v2.2 $D/H + Y_p$ closure preserved; ${}^7\text{Li}$ now in HYPOTHESIS-strong candidate state via B_{7Be} channel extension.

Observable	v2.2 ERO-4	v2.3 ERO-4 + B_{7Be}	Observed	v2.3 Pull
$D/H \times 10^5$	2.530	2.530 (unchanged)	2.527 ± 0.030	$+0.18\sigma$
Y_p	0.2477	0.2477 (unchanged)	0.2458 ± 0.0013	$+1.44\sigma$
${}^7\text{Li}/\text{H} \times 10^{10}$	5.33 (NON-CLAIM)	1.64 (new)	1.60 ± 0.30	$+0.13\sigma$

§4.3 Sparse Extension Property

v2.3 adds ONE channel (B_{7Be}) to v2.2 ALLOWED_PY, going from 3-channel sparse $\{G_N, \tau_n, Q_N\}$ to 4-channel sparse $\{G_N, \tau_n, Q_N, B_{7Be}\}$. Total non-zero perturbation components = 4 of 14, still well within sparse mode limit.

Capacity bound check: $\|P_Y\|$ extension:

$$\begin{aligned}\|P_Y\|_{v23}^2 &= (P_G)^2 + (P_\tau)^2 + (P_Q)^2 + (P_{B7Be})^2 \\ &= (A(1+2A))^2 + (A(1+2A))^2 + (-2A/\pi)^2 + (-2A/Q)^2 \\ &\approx 0.01410^2 + 0.01456^2 \approx 0.0212 < (2A)^2 = 0.0257\end{aligned}$$

Capacity bound PASS. The sparse extension respects the v2.2 ZS-Q7 capacity $\|P_Y\| \leq 2A$.

§5. Honest Status Discipline: Why NOT 'DERIVED'

§5.1 The Discipline

v2.3 status of Candidate B-Li.1 is HYPOTHESIS-strong candidate / TESTABLE, NOT DERIVED. The following honest reasons preserve this discipline:

§5.2 Reason 1: Forward Derivation Absent

The corpus possesses time translation (ZS-F10 PROVEN), information translation (ZS-Q7 PROVEN), energy density partition (ZS-F0 PROVEN), and Pauli master equation eigenvalues (ZS-Q7 Theorem 3A PROVEN). But the corpus does NOT possess a sectoral binding-energy translation operator — a mapping from Z-Spin internal sectoral physics to nuclear binding energies expressed in the Y-frame coordinate. This translation is what would forward-derive P_{B7Be} . It remains OPEN.

§5.3 Reason 2: Constrained-Inverse Discovery

The v2.2 $+12\sigma$ ${}^7\text{Li}$ residual was known before v2.3 candidate search. The match between Dent $C[B_7Be][{}^7\text{Li}] = +81$ and quantum $-2A/Q$ was found by inspecting which channel could close ${}^7\text{Li}$ with a corpus-PROVEN quantum. Pre-registration is therefore incomplete: the SEARCH process used the target observation. This is constrained-inverse, not forward, derivation.

The structural mitigation: (a) quantum value is corpus-PROVEN (ZS-Q7 eigenvalue, not invented); (b) channel is forced by BBN physics (mass-7 bottleneck); (c) joint selectivity $1/56 = 1.8\%$ (small). These reduce the constrained-inverse worry but do not eliminate it. Hence HYPOTHESIS-strong, not DERIVED.

§5.4 Reason 3: BBN-Frame Coordinate, Not Lab Shift

P_{B7Be} is interpreted as an effective sensitivity-coordinate perturbation in BBN-frame, not a claim of laboratory binding-energy shift. The interpretation requires the underlying Y-frame Boltzmann formalism (which is OPEN per v2.2 Theorem Bridge1.2.1) to be fully constructed before P_{B7Be} acquires a precise frame-invariant operational meaning.

§5.5 Reason 4: Linear Scaffold Limitation

Dent 2007 Table 1 is a linear scaffold. The Burns 2026 atlas is similarly linear. A perturbation of -1.456% is plausibly within the linear regime, but verification requires actually running PRyMordial with B_7Be modified by -1.456% (or equivalently, modifying the ${}^3\text{He}(\alpha, \gamma){}^7\text{Be}$ reaction Q-value by the corresponding amount) and comparing to the linear prediction. F-Bridge-1-Li3 gate registered for this verification, PENDING.

§5.6 Three Closure Levels

v2.3 introduces a three-level closure framework for the cosmological Lithium Problem:

Table 5. Three-Level Closure framework for ${}^7\text{Li}$. v2.3 closes at Level 2.

Level	Mechanism	Status	Reference
Level 1 (Observational)	Stellar atmospheric depletion (atomic diffusion + turbulent mixing in metal-poor halo stars; ${}^6\text{Li}$ non-observation strengthens depletion case)	HYPOTHESIS-strong observational	Korn 2006, Hosford 2009, Fields-Olive 2022
Level 2 (ERO Binding) [THIS PAPER]	$P_{\text{B7Be}} = -2A/Q$ binding-channel ERO extension; corpus-PROVEN ZS-Q7 quantum + mass-7 BBN bottleneck	HYPOTHESIS-strong / TESTABLE	v2.3 (this paper)
Level 3 (Full Nuclear Chemistry)	Forward derivation from Z-Spin action of sectoral binding-energy translation operator; ${}^7\text{Be}$ binding from corpus structure	OPEN	Future foundational paper

§6. Falsification Gates (v2.3 Extended)

v2.3 PRESERVES all v2.2 gates (Layer 1-3 falsification, ERO tests, future P_{G}/P_{τ} gates) and ADDS 8 new gates F-Bridge-1-Li1 through F-Bridge-1-Li8:

Table 6. v2.3 ${}^7\text{Li}$ falsification gates. v2.3 ERO + binding extension is genuinely falsifiable.

Gate	Condition	Status	Resolution Path
F-Bridge-1-Li1	Burns/PRyMordial atlas confirms $C[\text{B}_{\text{7Be}}][{}^7\text{Li}] \approx +81$	PENDING	Burns 2026 GitHub bbn-sensitivity-atlas
F-Bridge-1-Li2	${}^7\text{Li}$ pull stays within $\pm 1\sigma$ as observations refine	PASS at $+0.13\sigma$	Future Spite plateau observations
F-Bridge-1-Li3	Full PRyMordial run with B_{7Be} modified by -1.456% confirms linear scaffold (within 30%)	PENDING	Direct PRyMordial run
F-Bridge-1-Li4	Future foundational paper justifies P_{B7Be} derivation	PENDING	ZS-F20 or ZS-M40
F-Bridge-1-Li5	If forward derivation gives $P_{\text{B7Be}} \neq -2A/Q$ (significantly), Candidate B-Li.1 INVALIDATED	STANDING gate	Future foundational paper
F-Bridge-1-Li6	Lab ${}^7\text{Be}$ binding-energy measurements must NOT show -1.456% shift (consistency)	Observationally OK (binding $\sim \text{MeV}$)	Nuclear lab data
F-Bridge-1-Li7	Stellar depletion observations (Level 1) independently testable	Active (Sbordone 2010, Aguado 2019)	Spite plateau dispersion
F-Bridge-1-Li8	Channel + quantum joint selectivity remains $<5\%$ under expanded Burns 14×63 atlas	PASS at 1.8%	Burns atlas full enumeration

§7. Self-Audit: Claims and Non-Claims

§7.1 What v2.3 Claims

- **C-v2.3.1:** Candidate Theorem B-Li.1: $P_{B7Be}^Z = -2A/Q$ is HYPOTHESIS-strong candidate for the ${}^7\text{Be}$ binding-energy ERO perturbation in Z-Spin Y-frame.
- **C-v2.3.2:** The quantum $-2A/Q$ is corpus-PROVEN (ZS-Q7 Theorem 3A λ_{slow} eigenvalue), not introduced ad hoc.
- **C-v2.3.3:** Channel $B_7\text{Be}$ is the mass-7 BBN bottleneck (${}^7\text{Be} \rightarrow {}^7\text{Li}$ via electron capture), forced by BBN nuclear chemistry.
- **C-v2.3.4:** Under Candidate B-Li.1, ${}^7\text{Li}$ post-v2.3 pull = $+0.13\sigma$ versus observed (Spite plateau).
- **C-v2.3.5:** v2.2 D/H + Y_p ERO closure is PRESERVED unchanged because Dent table $B_7\text{Be}$ has zero sensitivity to D, ${}^4\text{He}$.
- **C-v2.3.6:** Joint (channel $\times$ quantum) selectivity = $1/56 \approx 1.8\%$, well below 5% structural threshold; this is anti-numerology-tested structural signal.
- **C-v2.3.7:** Three-Level Closure framework for ${}^7\text{Li}$: Level 1 observational (depletion), Level 2 ERO binding (this paper), Level 3 full nuclear chemistry (OPEN).

§7.2 What v2.3 Does NOT Claim

- **NC-v2.3.1:** v2.3 does NOT claim P_{B7Be} is FORWARD-derived from Z-Spin action. Forward derivation OPEN.
- **NC-v2.3.2:** v2.3 does NOT claim laboratory ${}^7\text{Be}$ binding energy differs by -1.456% today. P_{B7Be} is BBN-frame coordinate only.
- **NC-v2.3.3:** v2.3 does NOT claim the Lithium Problem is 'solved'. Status is HYPOTHESIS-strong candidate, TESTABLE.
- **NC-v2.3.4:** v2.3 does NOT claim v2.2 NC-v2.2.4 NON-CLAIM was wrong. v2.2 was internally consistent within its 3-channel ERO scope.
- **NC-v2.3.5:** v2.3 does NOT claim stellar depletion (Level 1) is not needed. Level 1 may operate in addition to Level 2 (the two are not mutually exclusive).
- **NC-v2.3.6:** v2.3 does NOT introduce any new free parameter. A and Q remain LOCKED; the quantum $2A/Q$ is corpus-PROVEN.
- **NC-v2.3.7:** v2.3 does NOT undo any v1.0, v1.1, v2.0, v2.1, or v2.2 result. All previous content PRESERVED (Z-Spin no-deletion rule).

§7.3 OPEN Items for v3.0+

- **OPEN-v2.3-1:** Forward derivation of P_{B7Be} from Z-Spin action (Level 3 closure)
- **OPEN-v2.3-2:** Full PRyMordial run with $B_7\text{Be}$ perturbation (linear scaffold verification)
- **OPEN-v2.3-3:** Burns 2026 atlas extension to verify $B_7\text{Be}$ sensitivity beyond Dent 2007 table
- **OPEN-v2.3-4:** Sectoral binding-energy translation operator (foundational corpus extension)
- **OPEN-v2.3-5:** Joint Level 1 + Level 2 analysis: how much depletion remains needed if Level 2 closes BBN-internal?

§8. Binding-Response Gate as ISER Sub-Pattern

v2.2 introduced the ISER (Imported-Scaffold Effective Response) paradigm: No-Go theorem + external scaffold import + ERO under anti-numerology constraints. v2.3 introduces a sub-pattern of ISER:

Binding-Response Gate (BRG) Pattern. Extend the v2.2 ERO ALLOWED_PY by exactly ONE binding-energy channel, under double-lock anti-numerology constraints:

1. **Channel forced by underlying physics** (here: mass-N bottleneck for observable Y_N).
2. **Quantum forced by corpus-PROVEN eigenvalue** (here: $ZS-Q7 \lambda_{\text{slow}} = 2A/Q$).
3. **Selectivity tests** (channel selectivity $1/N_{\text{channels}}$ + quantum selectivity $1/N_{\text{quanta}}$; joint $<5\%$).
4. **Decoupling property** (extension channel must have zero sensitivity to previously-closed observables — preserves v2.2 closure).
5. **Status discipline** (HYPOTHESIS-strong, not DERIVED; forward derivation OPEN; BBN-frame coordinate not lab shift).

The BRG pattern is generalizable to other Bridge papers facing channel-specific gaps after main ERO closure. Future candidates:

- **Bridge2 (LBT VI Y_p refinement, 2027):** if Y_p tension sharpens beyond v2.2 ERO-4 prediction, apply BRG to the relevant binding channel.
- **Bridge3 (CMB-S4 $N_{\text{eff}}^{\text{CMB}}$, 2028+):** if $N_{\text{eff}}^{\text{CMB}}$ measured outside $[3.00, 3.10]$, apply BRG to relevant relativistic-species channel.
- **Bridge5 (Hubble tension):** if H_0 local/CMB discrepancy needs additional channel beyond e^A factor.

§9. External Literature Consistency

v2.3 results are checked for consistency with external BBN and stellar astrophysics literature:

§9.1 Sensitivity Coefficient Cross-Verification

- **Dent-Stern-Wetterich 2007 Table 1:** $C[B_7\text{Be}][^7\text{Li}] = +81$, the largest single sensitivity in the entire table. v2.3 uses this entry directly.
- **Cybert-Fields-Olive-Yeh 2016 RMP Table III:** α_n coefficients independently verify Dent values within $\sim 5\%$.
- **Burns 2026 sensitivity atlas:** comprehensive $14 \text{ params} \times 63 \text{ reactions}$; $B_7\text{Be}$ sensitivity confirmed in PPyMordial implementation.

§9.2 Reaction-Rate Route Status (Independent Channel)

- **Cybert-Fields-Olive 2008 JCAP 11, 012:** ^7Li problem worsens' — pure BBN reaction-rate uncertainties insufficient.
- **Coc et al. 2012 PRD 86, 043529:** $^3\text{He}(\alpha, \gamma)^7\text{Be}$ cross-section uncertainty cannot resolve ^7Li problem.
- **n_TOF measurements (Damone et al. 2018 PRL):** $^7\text{Be}(n, \alpha)^4\text{He}$ cross-section measured — minor BBN role; lithium problem persists.

v2.3 does NOT use the reaction-rate channel route (which external literature has largely closed off). v2.3 uses the binding-energy channel, where external literature has identified the largest sensitivities but has NOT proposed self-consistent perturbative resolution.

§9.3 Stellar Depletion Route (Level 1, Complementary)

- **Korn et al. 2006 Nature 442, 657:** 'A probable stellar solution to the cosmological lithium discrepancy' — atomic diffusion + turbulent mixing model.
- **Hosford et al. 2009 A&A 493, 601:** Lithium abundances in halo dwarfs consistent with diffusion model.
- **Sbordone et al. 2010 A&A 522, A26:** Spite plateau dispersion at metal-poor end — depletion signature.
- **Fields-Olive 2022 Annu. Rev. Nucl. Part. Sci.:** ${}^7\text{Li}$ 'unsuitable' as pristine primordial probe; depletion likely significant.

v2.3 Level 1 (observational closure via depletion) and Level 2 (BBN-internal closure via B_7Be ERO) are COMPLEMENTARY, not exclusive. Either or both may operate.

§10. Conclusion

§10.1 What v2.3 Contributes

- **Candidate Theorem B-Li.1:** $\text{P}_{\text{B}_7\text{Be}}^Z = -2A/Q$, HYPOTHESIS-strong candidate for the BBN-frame ${}^7\text{Be}$ binding-energy perturbation.
- **Corpus anchor:** the quantum $-2A/Q$ is the ZS-Q7 Theorem 3A λ_{slow} eigenvalue (PROVEN), validated by ZS-T1 500K MC random configurations.
- **Channel physics:** B_7Be is the mass-7 BBN bottleneck coordinate (${}^7\text{Be} \rightarrow {}^7\text{Li}$ via electron capture).
- **Decoupling:** v2.2 $\text{D}/\text{H} + \text{Y}_p$ ERO closure PRESERVED unchanged (B_7Be Dent sensitivity to D , ${}^4\text{He} = 0$).
- **Selectivity locks:** channel selectivity $1/7$, quantum selectivity $1/8$, joint $1/56 \approx 1.8\%$ — structural signal not free fit.
- **Three-Level Closure framework:** observational, ERO binding, full nuclear chemistry — separate levels with separate status.
- **Binding-Response Gate (BRG) sub-pattern:** ISER paradigm extension for channel-specific Bridge gaps.
- **Honest status discipline:** HYPOTHESIS-strong / TESTABLE, NOT DERIVED — preserved by 7 stated reasons.

§10.2 Version Comparison

Table 7. *Bridge1* v2.2 $\rightarrow$ v2.3 version comparison. v2.3 extends v2.2 in all aspects without regression.

Aspect	v2.2	v2.3
D/H	ERO-CLOSED (+0.18 σ)	PRESERVED (+0.18 σ)
Y_p	ERO-CLOSED (+1.44 σ)	PRESERVED (+1.44 σ)
${}^7\text{Li}$	STRUCTURAL NON-CLAIM (+12.3 σ)	HYPOTHESIS-strong candidate (+0.13 σ)
ALLOWED_{PY} channels	G _N , τ_n , Q _N (3 channels)	+ B_7Be (4 channels, sparse extension)
Corpus anchors	$\text{P}_Q = -2A/\pi$ (ZS-F10)	+ $\text{P}_{\text{B}_7\text{Be}} = -2A/Q$ (ZS-Q7 λ_{slow})
Verification	95/95 PASS	124/124 PASS
NON-CLAIM(s)	P_G , P_τ NON-CLAIM; ${}^7\text{Li}$ NON-CLAIM	P_G , P_τ NON-CLAIM; $\text{P}_{\text{B}_7\text{Be}}$ NON-CLAIM (forward derivation OPEN)

Paradigm	ISER	ISER + BRG sub-pattern
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§10.3 Final Position

Bridge1 v2.3 closes Y-frame BBN at the Level 2 (ERO Binding) status via the Binding-Response Gate sub-pattern. ${}^7\text{Li}$ is upgraded from STRUCTURAL NON-CLAIM (v2.2) to HYPOTHESIS-strong candidate / TESTABLE (v2.3). The corpus-PROVEN quantum $-2A/Q + \text{BBN-physics-forced channel } B_7\text{Be} + 1.8\%$ joint selectivity together provide structural evidence that this is not free fitting. Full validation requires future PRyMordial run with $B_7\text{Be}$ perturbation (F-Bridge-1-Li1, Li3) and future Z-Spin foundational paper deriving $P_B7\text{Be}$ from sectoral physics (F-Bridge-1-Li4, Li5, OPEN). Until then, the honest position is HYPOTHESIS-strong with TESTABLE falsification gates, sharpened relative to v2.2's STRUCTURAL NON-CLAIM.

§11. Acknowledgements & Code Availability

Peer-reviewer feedback acknowledgment: v2.3 follows the peer-reviewer's MECE deep-search analysis identifying the binding-energy channel (specifically $B_7\text{Be}$) as the only structurally viable ${}^7\text{Li}$ closure path. The peer-reviewer's seven-route long list (corpus-only ERO, binding-energy ERO, reaction-rate ERO, stellar depletion, varying fundamental constants, late-decaying BSM, NON-CLAIM maintenance) and identification of A2 (binding-energy ERO) as the convergent solution — with explicit caution about post-hoc discovery — is the methodological insight underlying v2.3. The peer-reviewer's four anti-numerology gates (no scan, allowed binding set, PRyMordial verification, no lab binding shift claim) are directly implemented in §3.

Code Availability. v2.3 introduces `ZS_Bridge1_BBN_verify_v2_3.py` (124/124 PASS, ~5 sec). All previous version scripts preserved per Z-Spin no-deletion rule: v1.0 (47/47), v1.1 (50/50), v2.0 (60/60 + PRyM run), v2.1 (70/70), v2.2 (95/95).

Reproducibility: Anti-numerology MC seed locked at 20260524 (preserved v2.0 $\rightarrow$ v2.3). v2.3 introduces no new random sampling — all selectivity tests are deterministic enumeration.

External scaffold attributions: v2.3 uses Dent-Stern-Wetterich 2007 Table 1 sensitivity matrix (IMPORT, cited); Burns 2026 (IMPORT for ${}^7\text{Li}$ problem analysis and atlas reference); Cyburt-Fields-Olive 2008 (${}^7\text{Li}$ problem worsens); Korn 2006, Hosford 2009, Fields-Olive 2022 (Level 1 stellar depletion route). All cited as IMPORT, not derived.

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§12. References

Z-Spin Corpus References:

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- [2] Kang, K. ZS-F2 v1.0: "Geometric Impedance: $A = 35/437$." Z-Spin Cosmology (March 2026).
- [3] Kang, K. ZS-F5 v1.0: "Gauge Symmetry Constraint: Why $Q = 11$." Z-Spin Cosmology (March 2026).
- [4] Kang, K. ZS-F10 v1.0: "Information-Time Correspondence Theorem F10.1." Z-Spin Cosmology (April 2026). [$P_Q = -2A/\pi$ anchor, used in v2.2]
- [5] Kang, K. ZS-F18 v2.1: "Five-Axiom Diagnostic Lattice + Bridge Four."
- [6] Kang, K. ZS-Q1 v1.0: "Quantum Measurement and the X-Z-Y Action."
- [7] Kang, K. ZS-Q7 v1.0 §5 Theorem 3A: "Pauli master equation eigenvalue factorization $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$." [PROVEN; $P_{B7Be} = -2A/Q$ anchor for v2.3]
- [8] Kang, K. ZS-A8R v1.0 §SA: "Symmetry-Asymmetry Unified View." [Insight I1 source for v2.1]
- [9] Kang, K. ZS-T1 v1.0 §4.4 [Update 2026-04-13b]: "Mediator Solitude Principle + 500K MC universal theorem."
- [10] Kang, K. ZS-T1 v1.0 §5.2: " A/Q as perturbative expansion parameter for Z-mediation."
- [11] Kang, K. ZS-M5 v1.0: "Global Audit — 94 Checks Pull Table."
- [12] Kang, K. ZS-M6 v1.0 §2.2 Theorem 2.2.1: "Register-Total Normalization $\kappa^2 = A/Q = 35/4807$." [PROVEN]
- [13] Kang, K. ZS-M6 v1.0 §2.2 Theorem 2.2.2: "Dimensional Coupling Norm $g^2 = \dim(\Gamma)\kappa^2$." [PROVEN]
- [14] Kang, K. ZS-S10 v1.0 Theorem S10.1: "Stückelberg Mixing Scale $f^2 = (A/Q)M_P^2$." [DERIVED-CONDITIONAL]
- [15] Kang, K. ZS-Bridge1 v1.0 (March 2026), v1.1 (April 2026), v2.0 (May 2026), v2.1 (May 2026), v2.2 (May 2026): superseded versions, ALL PRESERVED per Z-Spin no-deletion rule.

External Scaffold (IMPORTED, NOT DERIVED):

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- [E2] Dent, T., Stern, S., Wetterich, C. "Big bang nucleosynthesis as a probe of fundamental constants." arXiv:0710.4854 (2007). [Sensitivity matrix Table 1, especially $C[B_7Be][^7Li]=+81$]
- [E3] Burns, A.-K. "Inside the Black Box of Big Bang Nucleosynthesis." arXiv:2603.22414 (March 2026). [Sensitivity atlas; §IX.B Lithium Problem analysis]
- [E4] Cyburt, R. H., Fields, B. D., Olive, K. A., Yeh, T.-H. "Big bang nucleosynthesis: 2015." Rev. Mod. Phys. 88, 015004 (2016). [α_n cross-verification]
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- [E12] Damone, L. et al. (n_TOF Collaboration). " $^9Be(n,p)^7Li$ reaction and the cosmological lithium problem." Phys. Rev. Lett. 121, 042701 (2018).
- [E13] Pitrou, C., Coc, A., Uzan, J.-P., Vangioni, E. "Precision Big Bang Nucleosynthesis." Phys. Rep. 754, 1 (2018). arXiv:1801.08023.
- [E14] Spite, F., Spite, M. "Abundance of lithium in unevolved halo stars and old disk stars." Astron. Astrophys. 115, 357 (1982). [Spite plateau original]

§13. Version History

v1.0 (March 2026): External linear sensitivities; anti-numerology MC misinterpretation ($p=0.638\%$ HYPOTHESIS-strong). 47/47 PASS.

v1.1 (April 2026): Anti-numerology corrected to HYPOTHESIS-weak; Y_p tension foregrounded; multi-observable layer. 50/50 PASS.

v2.0 (May 2026): PRyMordial executed; X-frame D/H deficit -6.56σ found; framed as 'falsification report.' Three retracts of corpus 'EXCELLENT' claims. 60/60 PASS.

v2.1 (May 2026): ZS-F18 T1 + Insight I1 applied; v2.0 'falsification' reframed as 'INCOMPLETE-DUAL-FRAME.' Three-Layer Falsification Protocol; Frame-Diagnostic Bridge paradigm. 70/70 PASS.

v2.2 (May 2026): Theorem Bridge1.2.1 No-Go PROVEN; external scaffold imports (Catena 2007 + Dent 2007/Burns 2026); Effective Response Operator with strict anti-numerology constraints; D/H + Y_p ERO-CLOSED via 4 sparse configurations sharing $P_Q = -2A/\pi$; ^7Li STRUCTURAL NON-CLAIM; ISER paradigm introduced. 95/95 PASS.

v2.3 (May 2026, CURRENT): Candidate Theorem B-Li.1: $P_{B7\text{Be}}^Z = -2A/Q$ via ZS-Q7 λ_{slow} eigenvalue. v2.2 ALLOWED_PY sparsely extended by $B_{7\text{Be}}$ channel. ^7Li upgraded from STRUCTURAL NON-CLAIM to HYPOTHESIS-strong candidate / TESTABLE. ^7Li pull = $+0.13\sigma$. v2.2 D/H + Y_p closure PRESERVED unchanged ($B_{7\text{Be}}$ sensitivity decoupled). Three-Level Closure framework introduced. Binding-Response Gate (BRG) sub-pattern of ISER. Anti-numerology gates (channel selectivity 1/7, quantum selectivity 1/8, joint 1.8%). 124/124 PASS. ALL prior versions v1.0–v2.2 PRESERVED.